

Comparing Geometric Brownian Motion and Dynamic Mode Decomposition For Financial Time Series Prediction

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Abstract—This paper explores two distinct approaches to modeling the stock market as a dynamical system, multivariate geometric Brownian motion (GBM) and dynamic mode decomposition (DMD), and compares how well they empirically perform at predicting short-term future states of the market. A historical data set of daily prices and returns for 527 U.S. stocks is used to benchmark these two approaches and produce a direct, meaningful comparison between the two, which to the best of the author’s knowledge has not been reported previously. Sensitivity of the prediction results to the size of the training data set used to fit the GBM and DMD models is examined. Sensitivity of the prediction results of the DMD approach to the DMD model complexity (rank) is also examined. The results suggest that, in this setting, and using root mean squared error and mean absolute error metrics to evaluate model performance, discovering dynamics from financial time series data using DMD yields better predictions overall than relying on the stochastic differential equations of (simple) multivariate GBM.

Index Terms—geometric Brownian motion, dynamic mode decomposition, dynamical systems, stock market, stock market prediction, financial time series

I. INTRODUCTION

In finance, the price of a stock is commonly modeled as a geometric Brownian motion (GBM). This model is very tractable and is compatible with standard efficient market theories in finance. However, it assumes a stationary process, normally distributed returns (percent changes in price), and continuity of the price path. Empirically, stock prices can exhibit non-stationary statistical properties, extreme values of returns occur at larger-than-normal frequencies, and price paths can jump discontinuously as traders respond to unexpected news. In other words, GBM cannot model market crashes and jumps, changing market conditions over time, or nonrandom effects of any external forces on the market, and may be too simple in its modeling of the random effects. This motivates other methods to model market dynamics for simulation or prediction purposes.

Dynamic mode decomposition (DMD) is a general and versatile method to discover dynamics from data. This paper compares the use of DMD versus use of an estimated GBM model for predicting short-term stock prices and returns based on many stocks’ historical time series data.

The primary advantage of DMD is that it is data-driven; it does not rely on pre-specified dynamical equations governing the system. It only assumes that a (time-invariant) transition matrix operates on the system state to yield an approximation of the next time step’s state, and that the input data matrix of system states over time has a low-rank structure. DMD obtains an estimate of the transition matrix as well as its eigenvalues and eigenvectors, modeling the system and estimating fundamental properties such as growth and decay modes and oscillation. The eigenvectors (dynamic modes) are approximations of a subset of the eigenfunctions of the infinite-dimensional linear Koopman operator. In Koopman theory for dynamical systems, this operator enables the representation of a nonlinear dynamical system as linear [1].

Also, the complexity of the model of the system obtained by DMD can be adjusted, unlike in the case of modeling as GBM. The maximum number of linearly independent dynamic modes to compute is a hyperparameter of the model, i.e., chosen before running the DMD algorithm.

Another key potential advantage of DMD is the improvement offered by one of its procedurally similar variants, dynamic mode decomposition with control (DMDc). Unlike original DMD, DMDc can account for the effect of external “forcing” inputs into the system when doing data-driven estimation of the dynamics. It can address nonrandom effects of external forces on the market, such as interest rates heavily influenced by central bank monetary policy (of which data are available), which GBM cannot. Implementing DMDc is beyond the scope of this study.

II. PRELIMINARIES

A. Geometric Brownian Motion

A Brownian (Wiener) process W_t is a continuous-time stochastic process whose change over a time interval Δt is normally distributed with mean 0 and variance Δt , i.e.,

$$\Delta W_t \triangleq W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t), \quad (1)$$

and has the following additional properties:

- $W_0 = 0$ almost surely,
- If $0 \leq s_1 < t_1 \leq s_2 < t_2$, then $W_{t_1} - W_{s_1}$ and $W_{t_2} - W_{s_2}$ are independent random variables,

- W_t is continuous in t almost surely. [2]

A stochastic process S_t follows a Geometric Brownian motion (GBM) if it satisfies the stochastic differential equation (SDE)

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (2)$$

where W_t is a Wiener process and μ and σ are constants. In financial modeling of a stock as S_t , the parameters μ and σ are referred to as “drift” and “volatility,” respectively.

In the multivariate case of GBM where there are N separate but correlated Wiener processes W_i , there are N SDEs:

$$dS_t^i = \mu_i S_t^i dt + \sigma_i S_t^i dW_t^i \text{ for } i = 1, \dots, N \quad (3)$$

where μ_i and σ_i are constants, and the (i, j) -element of the cross-sectional covariance matrix of the Wiener increments is

$$\mathbb{E} [dW_t^i dW_t^j] = \rho_{ij} \quad (4)$$

with $\rho_{ij} = 1$ for $i = j$. These $\rho_{ij} dt$ values are constant for all time t under the assumptions of pure GBM. When modeling a system of stocks as a multivariate GBM, the goal is to estimate from data the parameters μ_i , σ_i , and ρ_{ij} . In practice, the ultimate task is to simulate the system, so explicitly computing the individual σ_i and ρ_{ij} values is not necessary. It is sufficient to estimate functions of σ_i and ρ_{ij} that define appropriate probability distributions, which can then be sampled from to simulate a stochastic time evolution of the solution to (3).

B. Dynamic Mode Decomposition

If a (potentially nonlinear) dynamical system’s governing equations are unknown, DMD can be used to approximate the dynamics purely from snapshots corresponding to the state of the system measured at equispaced times. In the DMD procedure [3], we construct the approximate evolution of the system as linear through the operator $\mathbf{A}$:

$$\frac{d\mathbf{x}}{dt} \approx \mathbf{A}\mathbf{x}. \quad (5)$$

where state $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$. In terms of time-series data, the snapshot measured at the k th time, x_k , evolves to the next snapshot approximately as:

$$\mathbf{x}_{k+1} \approx \mathbf{A}\mathbf{x}_k. \quad (6)$$

Given a data set of m snapshots taken at equispaced times, we denote the sequence of snapshots collected at the first $m-1$ times,

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_{m-1}], \quad (7)$$

and the sequence of snapshots collected at the last $m-1$ times,

$$\mathbf{X}' = [\mathbf{x}_2 \quad \mathbf{x}_3 \quad \dots \quad \mathbf{x}_m]. \quad (8)$$

The matrix form of (6) describing the relationship between snapshots at every two consecutive times is

$$\mathbf{X}' \approx \mathbf{A}\mathbf{X}. \quad (9)$$

The system is thus

$$\mathbf{A} \approx \mathbf{X}'\mathbf{X}^+ \quad (10)$$

where $\mathbf{X}^+$ denotes the Moore-Penrose pseudoinverse of $\mathbf{X}$. The goal is to compute (10). The inputs of the DMD procedure are data matrices $\mathbf{X}$ and $\mathbf{X}'$, and the output is a particular computation of $\mathbf{A}$, which we will refer to as $\bar{\mathbf{A}}$, and the eigen-decomposition of $\bar{\mathbf{A}}$. This yields an approximate evolution of the system state, allowing for future state prediction given an initial state.

To compute the pseudoinverse $\mathbf{X}^+$, we utilize the singular value decomposition (SVD) of $\mathbf{X}$,

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^* \quad (11)$$

where $*$ denotes the complex conjugate transpose operator. In DMD, since we assume the data matrix $\mathbf{X}$ has a low rank structure, we actually choose a low rank (rank- r) approximation of $\mathbf{X}$ through a truncation of its SVD,

$$\mathbf{X}_{lowrank} = \tilde{\mathbf{U}}\tilde{\mathbf{\Sigma}}\tilde{\mathbf{V}}^* \approx \mathbf{X}, \quad (12)$$

by choosing a truncation value $r < m$ to retain only the r largest singular values and the corresponding left- and right-singular vectors of $\mathbf{X}$. Here, $\tilde{\mathbf{\Sigma}} \in \mathbb{R}^{r \times r}$ is the diagonal matrix of those r singular values in descending order, and $\tilde{\mathbf{U}} \in \mathbb{R}^{n \times r}$ and $\tilde{\mathbf{V}}^* \in \mathbb{R}^{r \times m-1}$ are the matrices containing the corresponding left- and right-singular vectors, respectively. This leads to computing a rank- r approximation for $\mathbf{X}^+$,

$$\mathbf{X}_{lowrank}^+ = \tilde{\mathbf{V}}\tilde{\mathbf{\Sigma}}^{-1}\tilde{\mathbf{U}}^* \approx \mathbf{X}^+. \quad (13)$$

Then (13) is plugged directly into (10), obtaining $\bar{\mathbf{A}}$ as a rank- r approximation of $\mathbf{A}$:

$$\bar{\mathbf{A}} = \mathbf{X}'\tilde{\mathbf{V}}\tilde{\mathbf{\Sigma}}^{-1}\tilde{\mathbf{U}}^* \quad (14)$$

Out of all rank- r matrices, $\bar{\mathbf{A}}$ is the least-squares solution to the problem $\mathbf{X}' = \mathbf{A}\mathbf{X}$ if it is overdetermined, or the minimum-norm solution to the problem $\mathbf{X}' = \mathbf{A}\mathbf{X}$ if it is underdetermined. This results from the Eckart-Young theorem [4] and the properties of the Moore-Penrose pseudoinverse [5]. Note the DMD procedure thus produces only up to r independent dynamic modes, so r is a tunable hyperparameter reflecting the complexity of the model obtained by DMD.

III. METHODOLOGY

A. Data

In finance, it’s of most practical interest to predict stock returns besides prices, since investment decisions are made not based on absolute asset prices but on returns on investment. A return is defined as the change in the total value of an investment in a stock over some period of time per monetary unit of initial investment.

To align with practical investment considerations while preserving methodological simplicity, this work addresses specific market data constraints. Firstly, in reality, certain corporate events on certain days, such as dividend payouts to shareholders, affect the listed price of a stock differently from the actual value of investing in the company. Therefore, calculation of returns needs to use “adjusted” prices that reflect the true total change in investment value (by accounting for the value of

dividend payouts each time they occur, for example). Such event information is recorded and available, so measurements of returns that account for these corrections are available.

Secondly, in reality, “daily” financial market data are only recorded for times when the markets are open, i.e., on trading days. These days exclude certain real days each year. In this work, trading day intervals are interpreted as no different from real day intervals. That is, trading days are assumed to align one-to-one with business days. This is slightly off from reality but is true for the most part (the vast majority of skipped calendar days are weekends). Firms generally do not work on non-business days and so do not generate value for shareholders over intervals between consecutive business days. Thus, returns calculated from listed prices and information on the stock’s event history, such as dividend payouts, are sufficient to approximate true daily returns on investment.

The data used in this work comes from the Daily Stock database from the Center for Research in Security Prices (CRSP), accessed through Wharton Research Data Services (WRDS) [6]. In this database, the return recorded on day T is the return for a sale on day T of the security (stock) purchased on the most recent day prior to T when the security had a valid price. Also, the returns in this database are calculated in a way that already accounts for events like dividend payouts. Data for U.S. stocks traded on U.S. exchanges were queried and obtained through WRDS. The data was cleaned to ensure the following conditions of the final data set:

- 1) Each security has a fully-recorded time series of returns that spans the full date range of trading days listed in the output of the database query, i.e., there are no dates in any security’s data range with a missing return,
- 2) Each security’s date range spans the maximum range among securities in the query output,
- 3) For each security, every daily return is computed based on the closing prices on the current and immediate preceding trading day, i.e., the data points are indeed all spaced equally apart in time by $\Delta t = 1$ trading day,
- 4) No two securities in the data set are issued for the same company merely with different share class designations. That is, if a company in the data set has more than one share class being traded during this date range, only one of its share classes was arbitrarily selected as the stock to represent it. Since such stocks are both directly shares of the same underlying firm value, they do not theoretically differ in their financial returns, and so are redundant components of the market state.

The final curated data set consists of the daily closing prices and daily returns of 527 securities, over 2012 total trading days (from January 3, 2017 through December 31, 2024).

When constructing the system state vector (dimension $N = 527$), a consistent ordering of the 527 stocks was chosen. The stocks are ordered by industry, and within each industry, ordered by decreasing market capitalization on January 3, 2017. A total of 206 unique industries are represented.

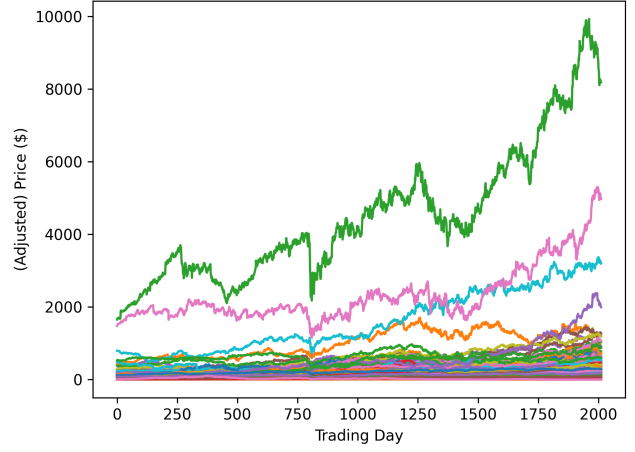


Fig. 1. Prices of all 527 stocks over all 2012 trading days from January 3, 2017 through December 31, 2024.

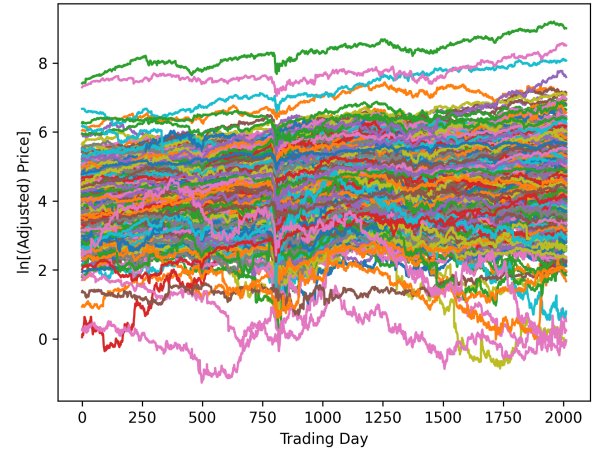


Fig. 2. Natural log of prices of all 527 stocks over all 2012 trading days from January 3, 2017 through December 31, 2024.

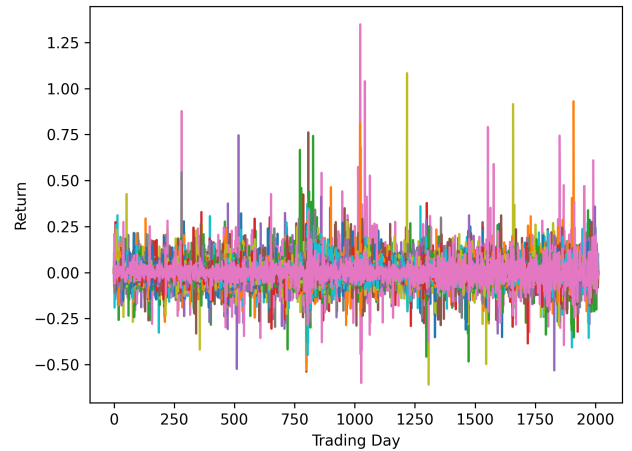


Fig. 3. Returns of all 527 stocks over all 2012 trading days from January 3, 2017 through December 31, 2024.

B. Train-Test Split of the Data and Performance/Fit Metrics

The first 1911 trading days are allocated as available for GBM/DMD model training. When training models, different subwindows of this data to actually fit the model to are tried:

- use all the available training data (1911 snapshots),
- use the final 600 snapshots in time before the test set,
- or use the final 100 snapshots in time before the test set. (For GBM, using only the final 100 training snapshots was not tried because this would cause the chosen parameter estimation method to fail.)

The remaining 101 days of the total data set thus comprise the test set on which to fairly compare GBM vs. DMD in terms of performance at predicting the short-term future. Specifically, only the final 100 days are used to compute model performance metrics, because the first day in the test set is utilized as an initial condition for the DMD model's predictions and for GBM price path simulation. For methodological consistency, the training data set is always treated similarly (i.e., the first day is excluded from computing model fit metrics on the training data).

Pooled root mean squared error (RMSE) and pooled mean absolute error (MAE) are the metrics used for comparing GBM and DMD models' goodness of fit to their training data and their goodness of prediction on test data. "Pooled" means the calculation involves averaging $N(K-1)$ error values, where N = number of stocks (527) and K = number of days in the data (sub)set on which the model is being evaluated. The $K-1$ is due to the fact that the first snapshot in the data window is always excluded from computing performance metrics. If x_{ik} denotes the i th component of the state vector $\mathbf{x}_k$ at the k th day in the data (sub)set, and $\hat{x}_{ik}$ denotes the model's prediction for this value, then

$$\text{RMSE} = \sqrt{\frac{\sum_{k=2}^K \sum_{i=1}^N (x_{ik} - \hat{x}_{ik})^2}{N(K-1)}} \quad (15)$$

$$\text{MAE} = \frac{\sum_{k=2}^K \sum_{i=1}^N |x_{ik} - \hat{x}_{ik}|}{N(K-1)}. \quad (16)$$

C. Multivariate GBM Model

1) *Parameter Estimation:* For a system of N stocks, suppose the i th stock's ("adjusted") price S_t^i is modeled as GBM according to (3). Dividing (3) through by S_t^i and also discretizing with equal-sized intervals Δt (representing an increment of one trading day, for example) gives a model for a stock's return over a given time interval $[t, t + \Delta t]$:

$$r_{t+\Delta t}^i \triangleq \frac{S_{t+\Delta t}^i - S_t^i}{S_t^i} = \frac{\Delta S_t^i}{S_t^i} \approx \mu_i \Delta t + \sigma_i \Delta W_t^i. \quad (17)$$

Here, $r_{t+\Delta t}^i$ is the holding period return of the i th stock at time $t + \Delta t$. It is the return of selling at time $t + \Delta t$ the stock i purchased at time t . For example, if Δt is one day, this is a daily return.

Denote the vectors

$$\epsilon_t \triangleq \begin{bmatrix} \sigma_1 \Delta W_t^{i=1} \\ \sigma_2 \Delta W_t^{i=2} \\ \vdots \\ \sigma_N \Delta W_t^{i=N} \end{bmatrix}, \quad \mu \triangleq \begin{bmatrix} \mu_{i=1} \\ \mu_{i=2} \\ \vdots \\ \mu_{i=N} \end{bmatrix}, \quad r_t \triangleq \begin{bmatrix} r_{t+\Delta t}^{i=1} \\ r_{t+\Delta t}^{i=2} \\ \vdots \\ r_{t+\Delta t}^{i=N} \end{bmatrix} \quad (18)$$

such that (17) can be written as the vector equation

$$r_{t+\Delta t} \approx \mu \Delta t + \epsilon_t. \quad (19)$$

From (1) and properties of Wiener process W_t^i , it follows that

$$\epsilon_t^i \triangleq \sigma_i \Delta W_t^i \sim \mathcal{N}(0, \sigma_i^2 \Delta t), \quad (20)$$

so ϵ_t follows a multivariate normal distribution with mean zero and variance-covariance matrix Ω (which in general is not a diagonal matrix):

$$\epsilon_t \sim \mathcal{N}(\mathbf{0}, \Omega). \quad (21)$$

From (4), the (i, j) element of Ω is a function of the volatilities of the i th and j th stocks as well as the correlation between their Wiener processes:

$$\Omega_{ij} \triangleq \mathbb{E} [\epsilon_t^i \epsilon_t^j] = \sigma_i \sigma_j \rho_{ij}. \quad (22)$$

Equations (19) and (21) imply that

$$r_{t+\Delta t} \sim \mathcal{N}(\mu \Delta t, \Omega). \quad (23)$$

The approximations in (17), (19), and (23) are increasingly accurate with smaller returns, i.e., with smaller Δt . On the scale of financial data spanning years, $\Delta t = 1$ trading day is small. In that case, (17) can be considered a GBM-based model for stock returns.

Moving forward, we set $\Delta t = 1$ trading day. Given a training data set $\{r_k\}_{k=1}^{K=K}$ of historical daily returns of N stocks for K consecutive trading days (where k denotes trading day index, not a time value), the goal is to estimate μ , the vector of drift parameters of all the stocks, and Ω , the cross-sectional variance-covariance matrix of all the stocks' returns. So long as Ω is positive definite, maximum likelihood estimation (MLE) of the parameters μ and Ω , treating (23) as exact, yields:

$$\hat{\mu} = \frac{1}{K} \sum_{k=1}^K r_k \quad (24)$$

$$\hat{\Omega} = \frac{1}{K} \sum_{k=1}^K (r_k - \hat{\mu})(r_k - \hat{\mu})^\top. \quad (25)$$

Note, MLE breaks down here when $N > K$, so the selected window size K for the utilized training data is restricted to be > 527 trading days.

2) *Simulation and Evaluation*: Given these parameter estimates, we have a full multivariate GBM model. Paths of daily returns of all stocks over time are then simulated by taking an independent sample from the multivariate normal distribution of (23) per day. The simulation's implied paths of the stocks' prices are then reconstructed using the definition of return in terms of price (the $\triangle$ relation in (17)) and using the empirical prices of the stocks on the initial day of the particular time window being simulated.

Goodness-of-fit of the estimated GBM model to both training data and test data is evaluated using the following Monte Carlo approach. For the training data set, 20 simulations are run. For each simulation, the pooled RMSE of the modeled paths of returns, as well as the implied paths of prices, is computed as (15). Then, the average RMSE value over the 20 simulations is computed. This average RMSE is the reported metric of model performance on the training data. The pooled mean absolute error (MAE), computed as (16), is also used in place of pooled RMSE. This entire evaluation procedure is done on the test data set as well.

D. DMD Models and Model Evaluation

Given a training data set, the two data matrices (7) and (8) are constructed as inputs to DMD, and the DMD algorithm implementation computes (14) following the procedure described in section II-B. A total of 12 different DMD models were fitted, varying along these dimensions:

- Different representations of the system state are tried: using stock returns as the data snapshots, vs. using prices as the data snapshots
- Different SVD truncation ranks (i.e., maximum number of modes estimated; value of r) are tried: 5, or 20
- Different training data windows are tried (either 100, 600, or 1911 daily state snapshots, as described in section III-B).

Goodness-of-fit of each DMD-discovered model to both training data and test data is evaluated by directly computing pooled RMSE and pooled MAE by (15) and (16).

IV. RESULTS AND DISCUSSION

Table I shows, for various GBM models trained on different subsets of the total available training data, the metrics for goodness of fit on the used training data set (excluding the first training data snapshot) and goodness of prediction on the 100 final days of test data. Tables II and III show these metrics for various DMD models utilizing different SVD truncation ranks (5 or 20) and trained on different subsets of the total available training data.

First, the sensitivity of model performances to the amount of training data used can be analyzed. For all models, training error (whether measured by RMSE or MAE) improves whenever fewer training data are used. In the case of stock prices as the system state, test error also improves whenever fewer training data are used. When stock returns represent the state, test error is improved when the last 600 training days are used rather than all 1911 training days. Conversely, when modeling

stock returns via DMD, test error worsens again when only the last 100 training days are used; this trend reversal is also more extreme for the model of rank-20 than rank-5. (This trend reversal is not seen if instead stock prices are modeled.)

The individual models' training error can be compared against their own test error to assess tendencies to overfit to training data and other behaviors. For the GBM model, regardless of the amount of data it was trained on, the fit of the simulated prices for the training set is always much worse than that for the test set (whether measured by RMSE or MAE). A similar observation applies to the case of simulating returns, although the difference is more marginal here. Neither overfitting nor underfitting is behind this phenomenon, however. The likely cause of the observation is that error accumulation is worse over a longer time horizon, such as 1911 or 600 training days versus only 100 test days. For the rank-5 and rank-20 DMD models, overfitting clearly occurs in all cases when stock prices represent the system state (and returns are not relevant), but when stock returns represent the state, overfitting seems to only happen when the smallest training data set is used. This suggests that DMD may generalize better when predicting returns rather than absolute prices.

The DMD models of the two different ranks can be compared against each other. Regardless of the size of the training data set window, the rank-20 model consistently achieves better training error than rank-5, regardless of if returns or prices represent the system state. But, when stocks' returns are used as the state, the rank-20 model consistently achieves worse test error than rank-5, and the opposite is seen when prices are used as the state. The optimal rank depends on whether the state comprises prices or returns.

Comparing the different models' test performances directly answers the research question. Regardless of if prices or returns represent the system state, both the rank-5 and rank-20 DMD models consistently show smaller error than the GBM model in both the test set as well as the training set (given the same training data window was used to fit the models). This result promisingly suggests that even the simplest version of the DMD algorithm offers a relative improvement to modeling and predicting short-term future states of the stock market compared to simple multivariate GBM.

TABLE I
GEOMETRIC BROWNIAN MOTION MODEL RESULTS

Stock Returns As System State				
	Average RMSE* on:		Average MAE* on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	0.03369	0.03231	0.02414	0.02320
Last 600 days	0.03192	0.03118	0.02301	0.02198
Stock Prices As System State				
	Average RMSE† on:		Average MAE† on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	389.04	96.44	92.00	28.56
Last 600 days	219.09	88.99	85.61	26.94

*Unitless. †In units of USD\$.

TABLE II
DMD-DISCOVERED RANK-5 MODEL RESULTS

Stock Returns As System State				
	RMSE* on:		MAE* on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	0.02367	0.02158	0.01504	0.01342
Last 600 days	0.02251	0.02138	0.01500	0.01321
Last 100 days	0.02155	0.02173	0.01338	0.01359
Stock Prices As System State				
	RMSE† on:		MAE† on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	17.21	46.63	9.22	23.62
Last 600 days	13.72	44.12	6.95	21.33
Last 100 days	8.59	41.32	3.65	17.25

*Unitless. †In units of USD\$.

TABLE III
DMD-DISCOVERED RANK-20 MODEL RESULTS

Stock Returns As System State				
	RMSE* on:		MAE* on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	0.02340	0.02185	0.01503	0.01363
Last 600 days	0.02217	0.02168	0.01488	0.01355
Last 100 days	0.01936	0.02271	0.01241	0.01444
Stock Prices As System State				
	RMSE† on:		MAE† on:	
Model trained on:	Train set	Test set	Train set	Test set
All 1911 days	7.96	32.77	4.28	17.44
Last 600 days	7.25	29.77	3.30	14.89
Last 100 days	6.03	40.71	1.98	16.42

*Unitless. †In units of USD\$.

V. LIMITATIONS AND FUTURE WORK

This work’s methodology can be improved in a few ways. Firstly, only a single GBM model is fitted to returns data, so to obtain simulated prices, they were derived from the GBM-simulated returns. An alternative is to estimate another GBM model directly on empirical price data and treated separately (just as how different DMD models in the current work are fit separately on price data and returns data). DMD models estimated from price data snapshots are comparable one-to-one against a GBM fitted on those exact same snapshots. Meanwhile, in the current methodology, the GBM and DMD results for the case of “stock prices as system state” (see the Tables) might not be totally equivalent to compare due to the fact that for GBM parameter estimation, price data snapshots were not actually used.

Secondly, the Monte Carlo approach to computing average performance metrics for the estimated GBM model used only 20 simulation runs to average over, which could be substantially too small to invoke the Central Limit Theorem and justify the obtained Average RMSE and Average MAE values. Thus, larger numbers of simulations should be run, to see whether using a large enough number leads to significantly different results and conclusions than what is presented in this paper.

Thirdly, rather than manually trying various hyperparameters like SVD truncation rank in DMD and training window

size, future work can instead utilize cross-validation techniques to tune hyperparameters.

Some further extensions can be considered. For example, besides geometric Brownian motion here, more advanced statistical and artificial-neural-network-based models currently studied and used by financial analysts can be explored and compared to DMD. Also, more robust versions of the DMD algorithm than the “Exact DMD” used here can be explored. DMD is not expected to effectively capture changing market conditions, i.e., when dynamics fundamentally become different in the test data than in the training data (and the “prices as system state” results show consistency with this expectation). But incorporating the ideas of other variants like online/streaming DMD could address this issue and further improve the results. Finally, future work can apply the DMDc algorithm incorporating time-series data of historical ‘exogenous’ forces on the market, such as central bank interest rate changes. DMDc will discover a model comprised of both ‘intrinsic’ market dynamics (A matrix) as well as a matrix describing the isolated effect of such exogenous forces (a B matrix). This work can explore how much the ‘intrinsic’ market dynamics discovered by DMDc differs from the dynamics discovered by DMD. The DMDc-discovered model is not very useful for short-term future prediction in practice, since this task requires having future exogenous forcing data in hand. Such data does not actually exist, and predicting such future forces would be a separate, much more complicated task (e.g., predicting pandemics that would prompt governments to make economic policy changes, and predicting how exactly governments would respond over time).

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